Decentralized Groupwise Expectation Propagation Detector for Uplink Massive MU-MIMO Systems

Hua Li, Yuanyuan Dong, Caihong Gong, Xiyuan Wang, *Member, IEEE*, and Xiaoming Dai, *Member, IEEE*

***Abstract*—With the proliferation of emerging Internet of Things (IoT) applications, massive multiuser multiple-input–multiple-output (MU-MIMO) is a promising technology to support the massive connectivity requirement with limited spectral resources. The existing detection algorithms are mainly designed based on a centralized baseband processing architecture, which requires extremely high raw baseband data rates transferred between base-station antennas and the central processing unit (CPU). Decentralized baseband processing (DBP) architecture has recently been proposed to alleviate high interconnect data rates and chip input/output bandwidth bottlenecks. Since the information is not fully shared among each antenna cluster, conventional decentralized detectors suffer from significant performance loss, especially for systems with high overload ratios and/or spatially correlated channels. In this work, we first propose a decentralized groupwise detection paradigm for the star architecture by dividing users into multiple user groups, which can be effectively exploited by factor graph-based message passing algorithms, such as the expectation propagation (EP) algorithm. Then, an efficient message fusion rule is devised based on the product principle of the multivariate Gaussian probability density function at the CPU. To reduce the computational complexity, an approximate groupwise EP (AGW-EP) method is proposed by judiciously selecting reliable constellation vectors during the symbol belief calculation phase. In addition, we extend the proposed methods to the daisy-chain architecture, which requires constant interconnect data rates. Simulation results demonstrate that the proposed groupwise paradigm greatly enhances the performance of the conventional EP detector. Moreover, the proposed decentralized groupwise EP and AGW-EP detection schemes outperform their counterparts, particularly in correlated MIMO channels.**

***Index Terms*—Decentralized baseband processing (DBP), expectation propagation (EP), groupwise, multiuser multiple-input–multiple-output (MU-MIMO).**

I. Introduction

MASSIVE connectivity is a central requirement for next-generation wireless systems that aim to support machine-type communications and the Internet of Things (IoT) [1], [2], [3]. According to Cisco, the number of IoT devices envisioned is expected to reach 500 billion by 2030 and will continue to increase exponentially, which imposes critical challenges for the load capacity of IoT networks [4], [5]. Massive multiuser multiple-input–multiple-output (MU-MIMO) is considered to be one of the main enablers in the fifth-generation (5G) network and beyond, due to its superiority in increasing spectral efficiency and enhancing link reliability [6], [7], [8]. Equipped with large-scale antennas, the base station (BS) can simultaneously provide service to tens or hundreds of users in the same time–frequency resource [9], [10]. However, most existing uplink detection algorithms are designed based on a centralized baseband processing architecture, which entails prohibitive computational complexity and excessively high raw baseband data rates transferred between BS radio-frequency (RF) units and the central processor. For a 128 BS-antenna massive MU-MIMO system operating at 80-MHz bandwidth with 10-bit analog-to-digital converters (ADCs), the raw baseband data rates approach 400 Gb/s [11]. Such high interconnect data rates exceed the existing common public radio interface (CPRI) [12], which is the industry standard for interface between remote radio units and baseband units. Additionally, the data rates scale linearly with the number of antennas, which imposes severe bottlenecks in terms of hardware implementation, chip input/output (I/O) bandwidth, and array scalability [13], [14]. Similar challenges persist for other increasingly popular technologies, including large intelligent surfaces (LISs) [15], [16] and cell-free massive MIMO [17], [18].

To solve the interconnection bottleneck, several different decentralized architectures have been studied in [13], [14], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], and [30]. In [13], a decentralized baseband processing (DBP) architecture has been proposed to reduce the front-haul data, in which the antenna array is partitioned into multiple independent clusters. Each cluster executes the baseband processing tasks (e.g., channel estimation and data detection) in an independent and parallel manner. Two distinct feedforward architectures, i.e., partially decentralized (PD) and fully decentralized (FD) architectures, have been investigated in [14] to overcome the achievable throughput limitation caused by the transfer latency of DBP. In [19] and [20], a decentralized daisy-chain architecture was designed, which allows data to be consumed close to where it is generated, thus saving energy and throughput. In addition, detailed analyses of

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the interconnection data rate, complexity, and latency were presented. Kulkarni et al. [21] proposed two hardware topologies, namely the ring and the star topologies, to discuss the throughput, latency, and interconnect bandwidth tradeoffs. The ring topology offers higher throughput at constant interconnect bandwidth, while the star topology yields lower latency. A generalized framework for centralized/decentralized architectures with WAX decomposition was proposed to evaluate the tradeoff between the level of decentralization and the associated requirement of interconnection bandwidth at the BS [22], [23].

Although DBP architectures can effectively alleviate the interconnect bandwidth bottleneck, their performance deteriorates dramatically in scenarios with a large number of antenna clusters [14], [24]. High-performance detectors and efficient fusion rules are of great importance in the potential application of DBP architectures. The decentralized maximum ratio combining (MRC), zero-forcing (ZF), and minimum mean-square error (MMSE) detectors were considered in [14], followed by the design of the optimal fusion rule. Unfortunately, the decentralized MRC provides poor spectral efficiency for systems with low BS-to-user antenna ratio configurations. The decentralized ZF and MMSE detectors suffer costly computational overheads at each cluster due to matrix inversion operations. To reduce the computational complexity, some inversion-free decentralized detectors, such as coordinate descent (CD) [25], conjugate gradient (CG), and alternating direction method of multipliers (ADMMs) have been proposed to approximate the performance of the MMSE detector [13], [26]. Zhao et al. [27] proposed a block CD (BCD)-based MMSE detection for the decentralized daisy chain architecture, which considers the practical colored noise scenario with a nondiagonal correlation matrix. In [28], a novel multistep fusion algorithm based on Gram matrix splitting was designed for the FD architecture to reduce the performance degradation caused by the weakened diagonal dominance of the Gram matrix in the practical finite-dimensional system. In [14] and [29], a nonlinear decentralized equalizer based on the large-MIMO approximate message passing (LAMA) was proposed for FD and PD architectures, which achieves individually optimal performance in the large-system limit. A decentralized expectation propagation (EP) detector, which approximates the symbol belief by a Gaussian probability density function (PDF) and exchanges the messages (i.e., means and variances) on a factor graph was presented in [24]. Zhang et al. [30] proposed a decentralized Gaussian message passing (GMP) receiver for the DBP architecture. In addition, the partition scheme of the antenna clusters and the convergence of the GMP method are also characterized by fixed-point analysis under the assumption of a large-system limit. Traditional decentralized detectors have mostly focused on the DBP architectures with high BS-to-user antenna ratios. However, considering the future need for massive connectivity in the cellular IoT network, a BS may be required to serve a large number of devices. The existing decentralized methods suffer from severe performance deterioration as the number of clusters increases. In addition, the vast majority of DBP-related literature assumes that the propagation channels are spatially uncorrelated. Although this assumption leads to analytical tractability, practical channels are generally spatially correlated in IoT networks due to the nonuniform radiation patterns of antennas [31]. The performance of decentralized detectors is affected by the spatial correlation of the channel to a large extent.

In this article, we propose a decentralized groupwise detection paradigm for uplink massive MU-MIMO systems with DBP architectures. In contrast to the conventional DBP-based approaches that perform interference cancelation on a user-by-user basis at each cluster, the proposed scheme suppresses the residual interference from all nondesired user groups, while fully exploiting the groupwise information of the desired group. As an illustrative example, the EP algorithm is employed in this work, which is an effective Bayesian computational approach for inferring intractable posterior densities. In the proposed scheme, the users are first divided into multiple groups and the groupwise EP (GW-EP) detection is carried out at the decentralized processor of each antenna cluster. Then, the output local messages are fused at the central processing unit (CPU) based on the judiciously designed message fusion rule. To reduce the computational complexity of the GW-EP detector, we propose an approximate GW-EP (AGW-EP) method, in which only a part of more reliable constellation vectors are selected to calculate the symbol beliefs at each iteration. The convergence of the proposed GW-EP algorithm is characterized by state evolution under the assumption of a large-system limit. In addition, we apply the proposed GW-EP/AGW-EP detectors, as well as the message fusion rule to the daisy-chain architecture. In this scheme, the a priori probability of the current cluster is calculated based on the local messages passed from the preceding cluster. The average interconnection data rate between nodes is independent of the number of antenna clusters for the daisy-chain architecture. Numerical results validate that the proposed GW-EP and AGW-EP detectors outperform their conventional counterparts in the star architecture, in particular for scenarios with a large number of clusters and spatially correlated channels. The proposed detectors in the daisy-chain architecture achieve comparable performance to the centralized architecture.

The remainder of this work is organized as follows. The uplink MU-MIMO system model and DBP architectures are described in Section II. In Section III, the detailed derivations of the proposed groupwise detectors are presented. Section IV provides the simulation results. Finally, we conclude this work in Section V.

Notation: Throughout this article, vectors and matrices are represented by lowercase boldface letters and uppercase boldface letters, respectively. The notations $(\cdot)^{\mathrm{T}}$, $(\cdot)^{-1}$, and $(\cdot)^{\mathrm{H}}$ denote the matrix transpose, inversion, and complex conjugate transpose, respectively. $\propto$ stands for equality up to scale. $\mathbb{C}^{N\times K}$ represents the set of $N \times K$ complex matrices. $\mathrm{diag}(\mathbf{x})$ forms a diagonal matrix with the elements of $\mathbf{x}$. $\mathbf{x}\backslash x$ expresses all the enumerations of $\mathbf{x}$ except for x. $\mathbf{I}_N$ denotes an identity matrix of size $N \times N$. $\mathbf{0}$ represents a vector with all elements being 0. ∞_N and $\mathbf{0}_N$ denote a matrix of size $N \times N$ with all elements being ∞ and 0, respectively. $\mathcal{CN}(\mathbf{x};\boldsymbol{\mu},\boldsymbol{\Omega}) = [1/\det(\pi\boldsymbol{\Omega})]e^{-(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{H}}\boldsymbol{\Omega}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$ represents a multivariate complex Gaussian distribution with mean vector

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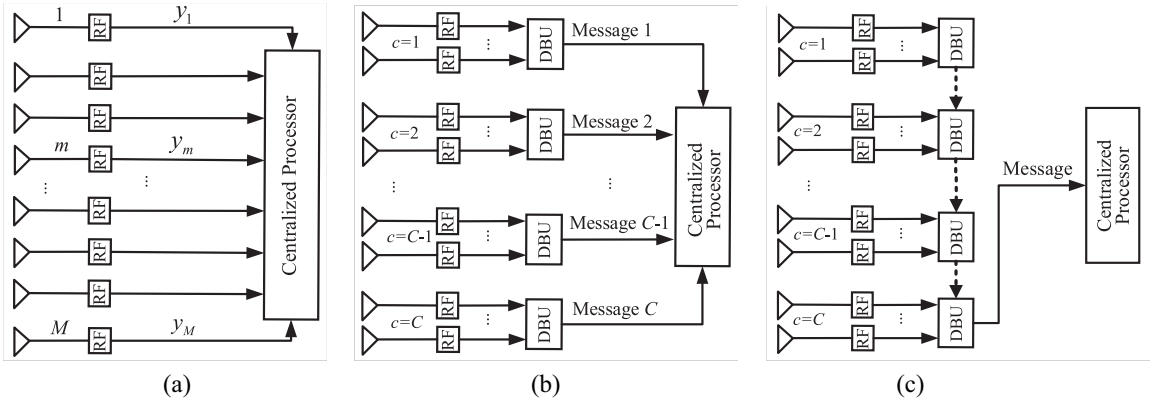


Fig. 1. Block diagram of centralized and DBP architectures. (a) Centralized architecture. The received signal at each antenna is forwarded to the centralized processor through its respective independent front-haul link. (b) Star architecture. Each DBU transmits the local messages to the centralized processor. (c) Daisy-chain architecture. Each DBU transmits the local messages to the next cluster, and only the last cluster has a direct connection with the centralized processor.

TABLE I
LIST OF MAIN VARIABLES AND THEIR PHYSICAL MEANINGS

Variable	Description
K	Number of single-antenna users
M	Number of antennas at the BS
$\mathcal{Q}$	The power-normalized QAM constellation set
C	Number of antenna clusters
N	Number of antennas of each antenna cluster
G	Number of user groups
K_g	Number of user in each group
$\mathbf{s} \in \mathbb{C}^K$	The transmit symbols vector of K users
$\mathbf{y}_c \in \mathbb{C}^N$	The received signals at the cth cluster
$\mathbf{H}_c \in \mathbb{C}^{N\times K}$	The local channel matrix at the cth cluster
$M^t_{\mathbf{x}_g\to f_{c,n}}$	The message propagated from $\mathbf{x}_g$ to $f_{c,n}$
$\mathbf{u}^t_{\mathbf{x}_g\to f_{c,n}}$, $\mathbf{V}^t_{\mathbf{x}_g\to f_{c,n}}$	The mean vector and covariance matrix of $M^t_{\mathbf{x}_g\to f_{c,n}}$
$M^t_{f_{c,n}\to\mathbf{x}_g}$	The message propagated from $f_{c,n}$ to $\mathbf{x}_g$
$u^t_{f_{c,n}\to\mathbf{x}_g}$, $V^t_{f_{c,n}\to\mathbf{x}_g}$	The mean and variance of $M^t_{f_{c,n}\to\mathbf{x}_g}$
$\beta^t_c(\mathbf{x}_g)$	The symbol belief of $\mathbf{x}_g$ at the cth cluster
$\alpha^t_c(\mathbf{x}_g)$	The multiplication of messages from all FNs to $\mathbf{x}_g$
$\mathbf{u}^t_{c,\mathbf{x}_g}$, $\mathbf{V}^t_{c,\mathbf{x}_g}$	The mean vector and covariance matrix of $\alpha^t_c(\mathbf{x}_g)$
$\widetilde{\mathbf{u}}^t_{c,\mathbf{x}_g}$, $\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_g}$	The posterior mean vector and covariance matrix
$\alpha(\mathbf{x}_g)$	The fused messages at the SN
$\gamma_c(\mathbf{x}_g)$	The messages passed from the cth cluster
$\boldsymbol{\mu}_{c,\mathbf{x}_g}$, $\boldsymbol{\Omega}_{c,\mathbf{x}_g}$	The mean vector and covariance matrix of $\gamma_c(\mathbf{x}_g)$

$\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Omega}$. For ease of reference, the main variables throughout this article are summarized in Table I, shown at the top of the page.

II. System Model

A. Uplink MU-MIMO System Model

Consider an uplink massive MU-MIMO system with K single-antenna users (IoT devices or sensors) and M receive-antennas ($M > K$) at the BS. The standard complex baseband relation between the transmitted and received signals is modeled as

$$\mathbf{y} = \mathbf{H}\mathbf{s} + \mathbf{w} \tag{1}$$

where $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_K] \in \mathbb{C}^{M\times K}$ represents the complex channel matrix, $\mathbf{s} = [s_1, \ldots, s_K]^{\mathrm{T}} \in \mathcal{Q}^K$ denotes the transmit symbol vector of K users, $\mathcal{Q}$ is the power-normalized quadrature amplitude modulation (QAM) constellation set with cardinality $|\mathcal{Q}| = Q$, and $\mathbf{w} \in \mathbb{C}^M$ is the additive white Gaussian noise (AWGN) vector with zero mean and covariance matrix $\sigma_n^2\mathbf{I}_M$.

B. Decentralized Baseband Processing Architectures

For the centralized architecture, the data/pilot signals received at all antennas are gathered at the centralized processor, which performs baseband processing tasks, e.g., synchronization, channel estimation, and data detection, as depicted in Fig. 1(a). In contrast, partial baseband processing tasks are moved from the CPU to the DBU in DBP architectures to overcome the interconnection bottleneck and reduce the computational complexity in the CPU. In this article, we consider two DBP architectures (i.e., star and daisy-chain architectures), which have been widely adopted in academia [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30]. For the star architecture in Fig. 1(b), all clusters are directly and unidirectionally connected to the centralized processor. Each cluster performs local baseband processing tasks in parallel. For the daisy-chain architecture in Fig. 1(c), all clusters are connected sequentially to each other by a dedicated connection and only the final cluster has a direct connection with the centralized processor.[1] Each cluster utilizes its local information and partial results from the prior cluster in the data detection phase. The DBP architectures partition the M BS antennas into C antenna clusters, where the cth cluster contains N antennas.[2] Each cluster is associated with an independent baseband processor. The channel matrix $\mathbf{H}$ is divided *rowwise* into C blocks as $\mathbf{H} = [\mathbf{H}_1^{\mathrm{T}}, \ldots, \mathbf{H}_C^{\mathrm{T}}]^{\mathrm{T}}$ with the local channel matrix $\mathbf{H}_c \in \mathbb{C}^{N\times K}$. The received signals $\mathbf{y}_c \in \mathbb{C}^N$at the cth cluster can be rewritten as

$$\mathbf{y}_c = \mathbf{H}_c\mathbf{s} + \mathbf{w}_c \tag{2}$$

where $\mathbf{w}_c \in \mathbb{C}^N$ represents the local AWGN vector.

[1]Without loss of generality, we assume that the order of antenna clusters is sequential, i.e., $c = 1, \ldots, C$.

[2]In this work, the BS antennas are assumed to be uniformly partitioned (i.e, $M = CN$) to minimize the chip I/O bandwidth.

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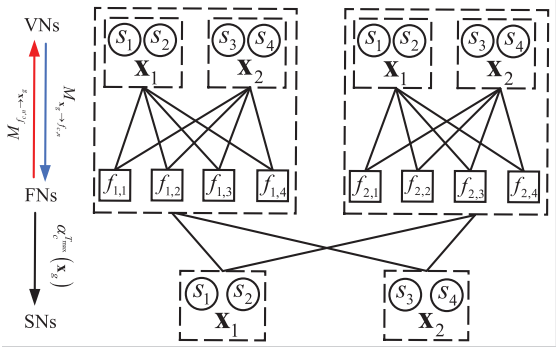


Fig. 2. Factor graph of the decentralized GW-EP detector for the star architecture with $M = 8$, $K = 4$, $G = 2$, and $C = 2$.

III. Proposed Decentralized Groupwise EP Detector

Since the information [e.g., the receive signals, noise samples, and channel state information (CSI)] is not fully shared among antenna clusters, conventional DBP-based detectors suffer from considerable performance loss, especially for DBP architectures with a large number of clusters [24]. We propose a groupwise scheme to improve the performance of the system with DBP architectures. The core idea of the groupwise scheme is to fully exploit the groupwise information and jointly suppress residual interferences for the users' signals in each user group. In the proposed groupwise detection scheme, we divide the users into G user groups.[3] The channel matrix at each cluster is partitioned *columnwise* into G group, e.g., $\mathbf{H}_c = [\mathbf{H}_{c,1}, \ldots, \mathbf{H}_{c,G}]$ with $\mathbf{H}_{c,g} \in \mathbb{C}^{N\times K_g}$. The corresponding input–output relation of the c cluster is expressed as

$$\mathbf{y}_c = \sum_g \mathbf{H}_{c,g}\mathbf{x}_g + \mathbf{w}_c \tag{3}$$

where $\mathbf{x}_g = [s_{(g-1)K_g+1}, \ldots, s_{gK_g}]^{\mathrm{T}} \in \mathbb{C}^{K_g}$ is the transmitted signal at the gth user group. As an illustrative example, we apply the groupwise scheme to the EP algorithm.

A. Decentralized GW-EP Detector for Star Architecture

1) GW-EP Detector: Without loss of generality, we take the cth cluster as an illustrative example. According to (3), the decentralized GW-EP detection of the uplink MIMO system can be represented by a factor graph as illustrated in Fig. 2. In the factor graph, variable nodes (VNs), factor nodes (FNs), and sum nodes (SNs) represent the user groups,[4] likelihood functions, and message fusions, respectively. The joint distribution $P(\mathbf{s}, \mathbf{y}_c)$ is factorized as [34]

$$\begin{aligned} P(\mathbf{s}, \mathbf{y}_c) &= P(\mathbf{s})P(\mathbf{y}_c \mid \mathbf{s}) \\ &= \prod_g P(\mathbf{x}_g) \prod_n P(y_{c,n} \mid \mathbf{x}_g). \end{aligned} \tag{4}$$

[3]Without loss of generality, the users are assumed to be uniformly divided, i.e., $K_1 = K_2 = \cdots = K_G$.

[4]For the factor graph-based methods, in contrast to the traditional EP algorithm that treats each user as a VN node, the proposed GW-EP method treats each user group as a VN node to make full use of the user groupwise information.

The likelihood function $P(y_{c,n} \mid \mathbf{x}_g)$ is given by

$$P(y_{c,n} \mid \mathbf{x}_g) \propto \exp\left(-\frac{1}{\sigma_n^2}\left|y_{c,n} - \mathbf{e}_{c,n}^{\mathrm{T}} \sum_g \mathbf{H}_{c,g}\mathbf{x}_g\right|^2\right) \tag{5}$$

where $\mathbf{e}_{c,n}$ denotes the nth column vector of $\mathbf{I}_N$. Let $M^t_{\mathbf{x}_g\to f_{c,n}}$ represent the message propagated from $\mathbf{x}_g$ to $f_{c,n}$ at the tth iteration ($t < T_{\max}$), and $M^t_{f_{c,n}\to\mathbf{x}_g}$ denote the message from the opposite direction. The message passing rules between VNs and FNs are given by [32]

$$M^t_{\mathbf{x}_g\to f_{c,n}}(\mathbf{x}_g) = P(\mathbf{x}_g) \prod_{n'\neq n} M^{t-1}_{f_{c,n'}\to\mathbf{x}_g}(\mathbf{x}_g) \tag{6}$$

$$M^t_{f_{c,n}\to\mathbf{x}_g}(\mathbf{x}_g) = \sum_{\mathbf{x}\setminus\mathbf{x}_g} P(y_{c,n} \mid \mathbf{x}_g) \prod_{g'\neq g} M^t_{\mathbf{x}_{g'}\to f_{c,n}}(\mathbf{x}_{g'}) \tag{7}$$

where $P(\mathbf{x}_g) = 1/Q^{K_g}$ is the a priori probability of $\mathbf{x}_g$. Since the symbol vectors take on values in the discrete constellation set $\mathcal{Q}$, the computational complexity of $M^t_{f_{c,n}\to\mathbf{x}_g}(\mathbf{x}_g)$ grows exponentially. To simplify the implementation of the sum-product algorithm, the concept of symbol belief based on EP is utilized to calculate approximate messages $M^t_{\mathbf{x}_g\to f_{c,n}}(\mathbf{x}_g)$ and $M^t_{f_{c,n}\to\mathbf{x}_g}(\mathbf{x}_g)$. EP is an algorithm for approximate statistical inference, which has extensive cutting-edge applications in multiuser detection, pattern recognition, and machine learning [24], [33], [34], [35], [36], [37]. The key idea of the EP algorithm is to approximate the intricate posterior distribution with a much simpler complex Gaussian distribution. Thus, the complicated symbol beliefs update can be replaced by mean vector and covariance matrix propagation [35]. The symbol belief (i.e., the approximate a posteriori probability) of $\mathbf{x}_g$ at the cth cluster is represented as

$$\beta^t_c(\mathbf{x}_g) = \frac{P_0(\mathbf{x}_g)\prod_n M^{t-1}_{f_{c,n}\to\mathbf{x}_g}(\mathbf{x}_g)}{\sum_{\mathbf{x}_g\in\mathcal{Q}^{K_g}} P_0(\mathbf{x}_g)\prod_n M^{t-1}_{f_{c,n}\to\mathbf{x}_g}(\mathbf{x}_g)} \tag{8}$$

where $P_0(\mathbf{x}_g) = 1/Q^{K_g}$ is the a priori probability of $\mathbf{x}_g$ at the cth cluster. We consider $\mathbf{x}_g$ as a vector of continuous random variables. Therefore, the message $M^t_{f_{c,n}\to\mathbf{x}_g}(\mathbf{x}_g)$ is approximated as a complex Gaussian PDF $M^t_{f_{c,n}\to\mathbf{x}_g}(\mathbf{x}_g) = \mathcal{CN}(\mathbf{H}_{c,g}\mathbf{x}_g; u^t_{f_{c,n}\to\mathbf{x}_g}, v^t_{f_{c,n}\to\mathbf{x}_g})$. The multiplicative message $\alpha^t_c(\mathbf{x}_g) = \prod_n M^{t-1}_{f_{c,n}\to\mathbf{x}_g}(\mathbf{x}_g)$ is assumed to follow a continuous multivariate complex Gaussian PDF $\alpha^t_c(\mathbf{x}_g) \propto \mathcal{CN}(\mathbf{x}_g; \mathbf{u}^t_{c,\mathbf{x}_g}, \mathbf{V}^t_{c,\mathbf{x}_g})$, where the covariance matrix and mean vector are computed, respectively, as

$$\begin{aligned} \left(\mathbf{V}^t_{c,\mathbf{x}_g}\right)^{-1} &= \sum_n \frac{1}{v^{t-1}_{f_{c,n}\to\mathbf{x}_g}} \mathbf{H}^{\mathrm{H}}_{c,g}\mathbf{e}_{c,n}\mathbf{e}^{\mathrm{T}}_{c,n}\mathbf{H}_{c,g} \\ &= \mathbf{H}^{\mathrm{H}}_{c,g}\Lambda^{t-1}_c\mathbf{H}_{c,g} \end{aligned} \tag{9}$$

$$\begin{aligned} \mathbf{u}^t_{c,\mathbf{x}_g} &= \mathbf{V}^t_{c,\mathbf{x}_g} \sum_n \frac{u^{t-1}_{f_{c,n}\to\mathbf{x}_g}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_g}} \mathbf{H}^{\mathrm{H}}_{c,g}\mathbf{e}_{c,n} \\ &= \mathbf{V}^t_{c,\mathbf{x}_g}\mathbf{H}^{\mathrm{H}}_{c,g}\Delta^{t-1}_c \end{aligned} \tag{10}$$

where $\Lambda^{t-1} = \mathrm{diag}(1/v^{t-1}_{f_{c,1}\to\mathbf{x}_g}, \ldots, 1/v^{t-1}_{f_{c,N}\to\mathbf{x}_g})$ and $\Delta^{t-1} = [u^{t-1}_{f_{c,1}\to\mathbf{x}_g}/v^{t-1}_{f_{c,1}\to\mathbf{x}_g}, \ldots, u^{t-1}_{f_{c,N}\to\mathbf{x}_g}/v^{t-1}_{f_{c,N}\to\mathbf{x}_g}]^{\mathrm{T}}$. The approximate

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symbol belief is obtained by minimizing the KL divergence $\mathrm{KL}(\beta_c^t(\mathbf{x}_g)\|\widetilde{\beta}_c^t(\mathbf{x}_g))$ [34], which obeys a continuous multivariate complex Gaussian PDF $\widetilde{\beta}_c^t(\mathbf{x}_g) = \mathcal{CN}(\mathbf{x}_g; \widetilde{\mathbf{u}}_{c,\mathbf{x}_g}^t, \widetilde{\mathbf{V}}_{c,\mathbf{x}_g}^t)$ with

$$\widetilde{\mathbf{u}}_{c,\mathbf{x}_g}^t = \mathbb{E}_{\beta_c^t(\mathbf{x}_g)}\left[\mathbf{x}_g\right] \tag{11}$$

$$\widetilde{\mathbf{V}}_{c,\mathbf{x}_g}^t = \mathbb{E}_{\beta_c^t(\mathbf{x}_g)}\left[\mathbf{x}_g\mathbf{x}_g^{\mathrm{H}}\right] - \widetilde{\mathbf{u}}_{c,\mathbf{x}_g}^t\left(\widetilde{\mathbf{u}}_{c,\mathbf{x}_g}^t\right)^{\mathrm{H}}. \tag{12}$$

Based on (6) and (8), the messages from VNs to FNs can be computed as

$$\begin{aligned} M_{\mathbf{x}_g\to f_{c,n}}^t(\mathbf{x}_g) &\propto \frac{\beta_c^t(\mathbf{x}_g)}{M_{f_{c,n}\to\mathbf{x}_g}^{t-1}(\mathbf{x}_g)} \\ &\approx \frac{\widetilde{\beta}_c^t(\mathbf{x}_g)}{M_{f_{c,n}\to\mathbf{x}_g}^{t-1}(\mathbf{x}_g)} \\ &= \mathcal{CN}\left(\mathbf{x}_g; \mathbf{u}_{\mathbf{x}_g\to f_{c,n}}^t, \mathbf{V}_{\mathbf{x}_g\to f_{c,n}}^t\right) \end{aligned} \tag{13}$$

where

$$\mathbf{V}_{\mathbf{x}_g\to f_{c,n}}^t = \left(\left(\widetilde{\mathbf{V}}_{c,\mathbf{x}_g}^t\right)^{-1} - \frac{\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{e}_{c,n}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}}{v_{f_{c,n}\to\mathbf{x}_g}^{t-1}}\right)^{-1} \tag{14}$$

$$\mathbf{u}_{\mathbf{x}_g\to f_{c,n}}^t = \mathbf{V}_{\mathbf{x}_g\to f_{c,n}}^t\left(\left(\widetilde{\mathbf{V}}_{c,\mathbf{x}_g}^t\right)^{-1}\widetilde{\mathbf{u}}_{c,\mathbf{x}_g}^t - \frac{u_{f_{c,n}\to\mathbf{x}_g}^{t-1}}{v_{f_{c,n}\to\mathbf{x}_g}^{t-1}}\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{e}_{c,n}\right). \tag{15}$$

Utilizing (5) and (7), we compute the mean and variance of $M_{f_{c,n}\to\mathbf{x}_g}^t(\mathbf{x}_g)$ from FNs to VNs as

$$u_{f_{c,n}\to\mathbf{x}_g}^t = y_{c,n} - \mathbf{e}_{c,n}^{\mathrm{T}}\sum_{g'\neq g}\mathbf{H}_{c,g'}\mathbf{u}_{\mathbf{x}_{g'}\to f_{c,n}}^t \tag{16}$$

$$v_{f_{c,n}\to\mathbf{x}_g}^t = \sigma_n^2 + \mathbf{e}_{c,n}^{\mathrm{T}}\left(\sum_{g'\neq g}\mathbf{H}_{c,g'}\mathbf{V}_{\mathbf{x}_{g'}\to f_{c,n}}^t\mathbf{H}_{c,g'}^{\mathrm{H}}\right)\mathbf{e}_{c,n}. \tag{17}$$

To avoid the matrix inversion in (14) and (15), by substituting (14) and (15) into (16) and (17), we rewrite the messages $u_{f_{c,n}\to\mathbf{x}_g}^t$ and $v_{f_{c,n}\to\mathbf{x}_g}^t$ as

$$u_{f_{c,n}\to\mathbf{x}_g}^t = y_{c,n} - \sum_{g'}\phi_{c,n,g'}^t\varphi_{c,n,g'}^t + \phi_{c,n,g}^t\varphi_{c,n,g}^t \tag{18}$$

$$v_{f_{c,n}\to\mathbf{x}_g}^t = \sigma_n^2 + \sum_{g'}\psi_{c,n,g'}^t\varphi_{c,n,g'}^t - \psi_{c,n,g}^t\varphi_{c,n,g}^t \tag{19}$$

where $\phi_{c,n,g}^t = \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}\widetilde{\mathbf{u}}_{c,\mathbf{x}_g}^t - (u_{f_{c,n}\to\mathbf{x}_g}^{t-1}/v_{f_{c,n}\to\mathbf{x}_g}^{t-1})\psi_{c,n,g}^t$, $\psi_{c,n,g}^t = \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}\widetilde{\mathbf{V}}_{c,\mathbf{x}_g}^t\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{e}_{c,n}$, and $\varphi_{c,n,g}^t = (v_{f_{c,n}\to\mathbf{x}_g}^{t-1}/[v_{f_{c,n}\to\mathbf{x}_g}^{t-1} - \psi_{c,n,g}^t])$ are all scalar quantities. The detailed derivation can be found in the Appendix.

2) Fusion of the Local Messages: When the maximum number of iterations is reached, the local messages of each cluster are fused at the CPU. The fused message of gth user group at SNs is computed based on the product principle of multivariate Gaussian PDF as

$$\begin{aligned} \alpha(\mathbf{x}_g) &\propto \prod_c \alpha_c^{T_{\max}}(\mathbf{x}_g) = \prod_c \mathcal{CN}\left(\mathbf{x}_g; \mathbf{u}_{c,\mathbf{x}_g}^{T_{\max}}, \mathbf{V}_{c,\mathbf{x}_g}^{T_{\max}}\right) \\ &= \mathcal{CN}\left(\mathbf{x}_g; \mathbf{u}_{\mathbf{x}_g}, \mathbf{V}_{\mathbf{x}_g}\right) \end{aligned} \tag{20}$$

Algorithm 1 Decentralized GW-EP Detector for the Star Architecture

Input: $\mathbf{H}$, $\mathbf{y}$, σ_n^2, $T_{\max}$
Initialization: $u_{f_{c,n}\to\mathbf{x}_g}^0 = 0$, $v_{f_{c,n}\to\mathbf{x}_g}^t = +\infty$, $P_0(\mathbf{x}_g) = 1/Q^{K_g}$
Each cluster carries out the GW-EP detection
for $t = 1, \ldots, T_{\max}$ **do**
  Compute $\alpha_c^t(\mathbf{x}_g)$ via (9) and (10).
  Compute $\beta_c^t(\mathbf{x}_g)$ via (8).
  Compute $\widetilde{\beta}_c^t(\mathbf{x}_g)$, $\widetilde{\mathbf{u}}_{c,\mathbf{x}_g}^t$, and $\widetilde{\mathbf{V}}_{c,\mathbf{x}_g}^t$ via (11) and (12).
  Compute $u_{f_{c,n}\to\mathbf{x}_g}^t$ and $v_{f_{c,n}\to\mathbf{x}_g}^t$ via (18) and (19).
end for
Fusion of the Decentralized Local Messages
Compute $\alpha(\mathbf{x}_g)$ via (20), (21), (22).
Compute the *a posteriori* probabilities $\beta(\mathbf{x}_g)$ via (20) and (23).
Compute the estimated soft symbol vector $\hat{\mathbf{x}}_g$ via (24).
Output: $\hat{\mathbf{x}}_g$

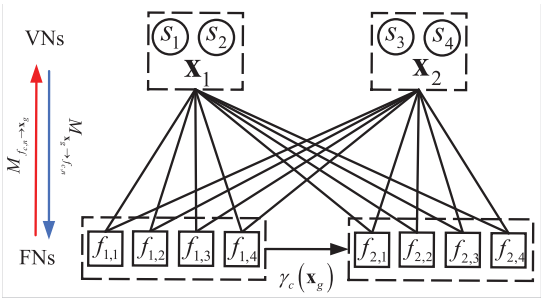


Fig. 3. Factor graph of the decentralized GW-EP detector for the daisy-chain architecture with $M = 8$, $K = 4$, $G = 2$, and $C = 2$.

where the covariance matrix and mean vector are given, respectively, by

$$\mathbf{V}_{\mathbf{x}_g} = \left(\sum_c\left(\mathbf{V}_{c,\mathbf{x}_g}^{T_{\max}}\right)^{-1}\right)^{-1} \tag{21}$$

$$\mathbf{u}_{\mathbf{x}_g} = \mathbf{V}_{\mathbf{x}_g}\left(\sum_c\left(\mathbf{V}_{c,\mathbf{x}_g}^{T_{\max}}\right)^{-1}\mathbf{u}_{c,\mathbf{x}_g}^{T_{\max}}\right). \tag{22}$$

The symbol belief at the CPU can be calculated as

$$\beta(\mathbf{x}_g = \boldsymbol{\xi}) = \frac{P_0(\mathbf{x}_g = \boldsymbol{\xi})\alpha(\mathbf{x}_g = \boldsymbol{\xi})}{\sum_{\boldsymbol{\xi}'\in\mathcal{Q}^{K_g}} P_0(\mathbf{x}_g = \boldsymbol{\xi}')\alpha(\mathbf{x}_g = \boldsymbol{\xi}')}. \tag{23}$$

Finally, the estimated soft symbol vector of the gth group is computed as

$$\hat{\mathbf{x}}_g = \sum_{\boldsymbol{\xi}\in\mathcal{Q}^{K_g}}\boldsymbol{\xi}\beta(\mathbf{x}_g = \boldsymbol{\xi}). \tag{24}$$

The proposed decentralized GW-EP detector for the star architecture is summarized in Algorithm 1.

B. Decentralized GW-EP Detector for the Daisy-Chain Architecture

1) GW-EP Detector: The factor graph of the proposed GW-EP method for the daisy-chain architecture is illustrated in Fig. 3. Each cluster computes partial information locally, fuses

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it with received local messages from the preceding cluster, and then forwards the combined messages to the next cluster. Based on the mean vectors and covariance matrices of the multivariate Gaussian PDF passed from the preceding cluster, the a priori probability is first calculated at each cluster as

$$P_c(\mathbf{x}_g) = \frac{P_0(\mathbf{x}_g)\gamma_{c-1}(\mathbf{x}_g)}{\sum_{\mathbf{x}_g \in \mathcal{Q}^{K_g}} P_0(\mathbf{x}_g)\gamma_{c-1}(\mathbf{x}_g)} \tag{25}$$

where $P_0(\mathbf{x}_g) = 1/Q^{K_g}$ is the initial a priori probability of all clusters, $\gamma_0 = \mathcal{CN}(\mathbf{x}_g; \mu_{0,\mathbf{x}_g}, \Omega_{0,\mathbf{x}_g})$ with $\boldsymbol{\mu}_{0,\mathbf{x}_g} = \mathbf{0}$ and $\boldsymbol{\Omega}_{0,\mathbf{x}_g} = +\infty_{K_g}$. At each iteration of the GW-EP detector, the symbol belief of $\mathbf{x}_g$ is calculated as

$$\begin{aligned}\beta_c^t(\mathbf{x}_g) &= \frac{P_c(\mathbf{x}_g)\prod_n M_{f_{c,n}\to\mathbf{x}_g}^{t-1}(\mathbf{x}_g)}{\sum_{\mathbf{x}_g \in \mathcal{Q}^{K_g}} P_c(\mathbf{x}_g)\prod_n M_{f_{c,n}\to\mathbf{x}_g}^{t-1}(\mathbf{x}_g)} \\ &= \frac{P_c(\mathbf{x}_g)\alpha_c^t(\mathbf{x}_g)}{\sum_{\mathbf{x}_g \in \mathcal{Q}^{K_g}} P_c(\mathbf{x}_g)\alpha_c^t(\mathbf{x}_g)}\end{aligned} \tag{26}$$

where the mean vector and covariance matrix of $\alpha_c^t(\mathbf{x}_g)$ are computed based on (9) and (10), respectively.

2) Fusion of the Local Messages: When the maximum number of iterations is reached, the local messages $\alpha_c^{T_{\max}}(\mathbf{x}_g)$ of the current cluster and the messages $\gamma_{c-1}(\mathbf{x}_g)$ passed from the preceding cluster are fused at the current DBU

$$\begin{aligned}\gamma_c(\mathbf{x}_g) &= \alpha_c^{T_{\max}}(\mathbf{x}_g)\gamma_{c-1}(\mathbf{x}_g) \\ &= \mathcal{CN}(\mathbf{x}_g; \boldsymbol{\mu}_{c,\mathbf{x}_g}, \boldsymbol{\Omega}_{c,\mathbf{x}_g})\end{aligned} \tag{27}$$

where the combined covariance matrix and mean vector are given, respectively, by

$$\boldsymbol{\Omega}_{c,\mathbf{x}_g} = \left(\left(\mathbf{V}_{c,\mathbf{x}_g}^{T_{\max}}\right)^{-1} + \left(\boldsymbol{\Omega}_{c-1,\mathbf{x}_g}\right)^{-1}\right)^{-1} \tag{28}$$

$$\boldsymbol{\mu}_{c,\mathbf{x}_g} = \boldsymbol{\Omega}_{c,\mathbf{x}_g}\left(\left(\mathbf{V}_{c,\mathbf{x}_g}^{T_{\max}}\right)^{-1}\mathbf{u}_{c,\mathbf{x}_g}^{T_{\max}} + \left(\boldsymbol{\Omega}_{c-1,\mathbf{x}_g}\right)^{-1}\boldsymbol{\mu}_{c-1,\mathbf{x}_g}\right). \tag{29}$$

Finally, the last cluster transmits its messages to the CPU, and the symbol belief $\beta(\mathbf{x}_g)$ of the gth group can be calculated based on (25) by replacing $\gamma_{c-1}(\mathbf{x}_g)$ with $\gamma_C(\mathbf{x}_g)$ as

$$\beta(\mathbf{x}_g) = \frac{P_0(\mathbf{x}_g)\gamma_C(\mathbf{x}_g)}{\sum_{\mathbf{x}_g \in \mathcal{Q}^{K_g}} P_0(\mathbf{x}_g)\gamma_C(\mathbf{x}_g)}. \tag{30}$$

The soft symbol vector of the gth group can be computed based on (24). The proposed decentralized GW-EP detector for the daisy-chain architecture is summarized in Algorithm 2.

C. Approximate Groupwise EP Detector

With a closer observation of (8) and (26), we find that the main computational complexity of the GW-EP algorithm lies in the calculation of symbol beliefs, which depends on the modulation order Q and the number of users K_g per group. To reduce the computational complexity, we propose an AGW-EP method by judiciously selecting the most reliable constellation vectors (i.e., with larger symbol beliefs) according to symbol beliefs in the last iteration $\beta_c^{t-1}(\mathbf{x}_g)$. The number of reliable constellation vectors is set as $K_g Q$ for the AGW-EP method, thus exhibiting the same order of computational complexity as the conventional EP detector. In fact, the performance loss due to the choice of $K_g Q$ is marginal for both DBP architectures, as demonstrated later in Section IV.

Algorithm 2 Decentralized GW-EP Detector for the Daisy-Chain Architecture

Input: $\mathbf{H}$, $\mathbf{y}$, σ_n^2, $T_{\max}$
Initialization: $u_{f_{c,n}\to\mathbf{x}_g}^0 = 0$, $v_{f_{c,n}\to\mathbf{x}_g}^t = +\infty$, $P_0(\mathbf{x}_g) = 1/Q^{K_g}$, $\boldsymbol{\mu}_{0,\mathbf{x}_g} = \mathbf{0}$, $\boldsymbol{\Omega}_{0,\mathbf{x}_g} = +\infty_{K_g}$
for $c = 1, \ldots, C$ **do**
  Compute the *a priori* probability $P_c(\mathbf{x}_g)$ via (25).
  for $t = 1, \cdots, T_{\max}$ **do**
    Compute $\alpha_c^t(\mathbf{x}_g)$ via (9) and (10).
    Compute $\beta_c^t(\mathbf{x}_g)$ via (26).
    Compute $\widetilde{\beta}_c^t(\mathbf{x}_g)$, $\widetilde{\mathbf{u}}_{c,\mathbf{x}_g}^t$, and $\widetilde{\mathbf{V}}_{c,\mathbf{x}_g}^t$ via (11) and (12).
    Compute $u_{f_{c,n}\to\mathbf{x}_g}^t$ and $v_{f_{c,n}\to\mathbf{x}_g}^t$ via (18) and (19).
  end for
  Fusion of the Decentralized Local Messages
  Compute $\gamma_c(\mathbf{x}_g)$ via (27), (28), (29).
end for
Compute the estimated soft symbol vector $\hat{\mathbf{x}}_g$ via (24).
Output: $\hat{\mathbf{x}}_g$

Remark 1: For the daisy-chain architecture, the computational complexity of $\beta_c^t(\mathbf{x}_g)$ can be further reduced since the latter cluster can exploit the a priori information (i.e., a posteriori output) of the preceding cluster.

D. Convergence Analysis

The convergence of the proposed decentralized GW-EP algorithm is characterized by the state evolution, which derives the distribution of messages at the $(t+1)$ th iteration according to that of the tth iteration [36], [37]. Based on the large system limit (i.e., $K \to +\infty$, $N \to +\infty$, and $\lambda = N/K \geq 1$ being a fixed constant), we assume $\mathbf{V}_{\mathbf{x}\to f_c} = \lim_{t\to+\infty} \mathbf{V}_{\mathbf{x}_g\to f_{c,n}}^t$, $v_{f_c\to\mathbf{x}} = \lim_{t\to+\infty} v_{f_{c,n}\to\mathbf{x}_g}^t$, and $\widetilde{\mathbf{V}}_{c,\mathbf{x}} = \lim_{t\to+\infty} \widetilde{\mathbf{V}}_{c,\mathbf{x}_g}^t$ for $\forall n \in [1, \ldots, N]$ and $\forall g \in [1, \ldots, G]$.

Proposition 1: At the VNs, the output is given by

$$\mathbf{V}_{\mathbf{x}\to f_c} \approx \widetilde{\mathbf{V}}_{c,\mathbf{x}}. \tag{31}$$

Proof: By submitting (17) into (14), we have

$$\begin{aligned}\mathbf{V}_{\mathbf{x}\to f_c} &= \left(\left(\widetilde{\mathbf{V}}_{c,\mathbf{x}}\right)^{-1} - \frac{\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{e}_{c,n}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}}{\sigma_n^2 + \mathbf{e}_{c,n}^{\mathrm{T}}\left(\sum_{g'\neq g}\mathbf{H}_{c,g'}\mathbf{V}_{\mathbf{x}\to f_c}\mathbf{H}_{c,g'}^{\mathrm{H}}\right)\mathbf{e}_{c,n}}\right)^{-1} \\ &= \left(\left(\widetilde{\mathbf{V}}_{c,\mathbf{x}}\right)^{-1} - \frac{\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{e}_{c,n}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}}{\sigma_n^2 + (G-1)\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\mathbf{V}_{\mathbf{x}\to f_c}\mathbf{H}_{c,g'}^{\mathrm{H}}\mathbf{e}_{c,n}}\right)^{-1} \overset{(a)}{\approx} \widetilde{\mathbf{V}}_{c,\mathbf{x}}\end{aligned} \tag{32}$$

where (a) holds due to the fact that the second term approaches zero when $G = K/K_g \to +\infty$. ■

Proposition 2: At the FNs, we have

$$\mathbf{V}_{c,\mathbf{x}} \approx \frac{\sigma_n^2 + G\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}\mathbf{V}_{\mathbf{x}\to f_c}\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{e}_{c,n}}{N}\mathbf{I}_{K_g} \tag{33}$$

where $\mathbf{V}_{c,\mathbf{x}} = \lim_{t\to+\infty} \mathbf{V}_{c,\mathbf{x}_g}$ for $\forall g \in [1, \ldots, G]$.

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TABLE II
COMPUTATIONAL COMPLEXITY

CG	$(T-1)(8CNK+10CK+2K)+12CNK+4CK+4K$
MRC	$C(K^3+3K^2+5K)+NK^2+NK$
MMSE	$C(\frac{10}{3}K^3+\frac{7}{2}K^2+2K)+NK^2+NK$
LAMA	$CT_{max}(6KQ+(K+1)^2)+C(NK^2+NK+4K+2)+N$
GMP/EP	$CT_{max}K(6N+5Q+4)+4QK+2CK+2K$
GW-EP	$T_{max}CG(Q^{K_g}(\frac{5}{2}K_g^2+\frac{1}{2}K_g)+3K_g^2N+3K_gN+4)+2(C+1)GK_g^2+GQ^{K_g}(K_g+1)^2$
AGW-EP	$T_{max}CG(QK_g(\frac{5}{2}K_g^2+\frac{1}{2}K_g)+3K_g^2N+3K_gN+4)+CGQ^{K_g}(\frac{5}{2}K_g^2+\frac{1}{2}K_g)+2(C+1)GK_g^2+GQ^{K_g}(K_g+1)^2$

Proof: By submitting (17) into (9), we have

$$\begin{aligned}\mathbf{V}_{c,\mathbf{x}} &= \left(\frac{\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{H}_{c,g}}{v_{f_c\to\mathbf{x}}}\right)^{-1} = v_{f_c\to\mathbf{x}}\left(\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{H}_{c,g}\right)^{-1}\\ &= \left(\sigma_n^2+G\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}\mathbf{V}_{\mathbf{x}\to f_c}\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{e}_{c,n}\right)\left(\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{H}_{c,g}\right)^{-1}\\ &\approx \frac{\sigma_n^2+G\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}\mathbf{V}_{\mathbf{x}\to f_c}\mathbf{H}_{c,g}^{\mathrm{H}}\mathbf{e}_{c,n}}{N}\mathbf{I}_{K_g}. \end{aligned} \tag{34}$$

■

Remark 2: When the symbol belief [cf. (8)] for a certain element converges to 1, i.e., $\beta_c^t(\mathbf{x}_g=\boldsymbol{\xi})=1$, we obtain $\mathbf{V}_{\mathbf{x}\to f_c}\approx\widetilde{\mathbf{V}}_{c,\mathbf{x}}\to\mathbf{0}_{K_g}$ according to (11), (12), and (32). Therefore, $v_{f_c\to\mathbf{x}}\to\sigma_n^2$ is obtained based on (17). The gap between $v_{f_c\to\mathbf{x}}$ and σ_n^2 can be used to evaluate the performance of the proposed detectors. In Section IV, the state evolution of $v_{f_c\to\mathbf{x}}$ will be tracked to validate the convergence and the performance of the proposed detectors. The bit error rate (BER) performance is uniquely determined at the fixed-point as the iteration increases [38].

E. Computational Complexity Analysis

The computational complexity is assessed in terms of the number of complex multiplications [39]. The computational complexity of the decentralized GW-EP detector can be divided into three parts, i.e., the calculation of messages between FNs and VNs: $T_{\max}CG(Q^{K_g}([5/2]K_g^2+[1/2]K_g)+3K_g^2N+3K_gN+4)$, the calculation of fusion of the decentralized message: $2(C+1)GK_g^2$, and the computation of the a posteriori probabilities at VNs: $GQ^{K_g}(K_g+1)^2$. In addition, the computational complexity of the MRC, CG [26], MMSE [14], GMP [30], and LAMA [29] detectors is also provided in Table II.

F. Interconnect Data Rates

The average interconnect data rates of the proposed detectors are measured in terms of the number of real symbols entered into the CPU. For the centralized architecture, each antenna forwards the received data signal y_m to the CPU, which corresponds to two real symbols. This amounts to $2M$ real symbols being transmitted from all antennas to the CPU, which is proportionate to the number of BS antennas. For the star architecture, each antenna cluster sends $\mathbf{u}_{c,\mathbf{x}_g}^{T_{\max}}$ and $\mathbf{V}_{c,\mathbf{x}_g}^{T_{\max}}$ to the CPU, where $g\in[1,\ldots,G]$. The total number of real symbols amounts to $CK(2+K_g)$. For the daisy-chain architecture, the messages $\boldsymbol{\mu}_{c,\mathbf{x}_g}$ and $\boldsymbol{\Omega}_{c,\mathbf{x}_g}$ are passed through the nodes (DBU or CPU) sequentially. The number of real symbols entered into the CPU is $K(2+K_g)$, which is independent of the number of antenna clusters.

IV. SIMULATION RESULTS

In this section, simulation results are presented to verify the validity of the proposed decentralized GW-EP/AGW-EP detectors for the uplink massive MU-MIMO system.

A. Simulations Parameters

We consider two different channel models, i.e., uncorrelated and correlated Rayleigh fading channels. For the uncorrelated Rayleigh fading channel, $\mathbf{h}_k$ is distributed as $\mathbf{h}_k\sim\mathcal{CN}(0,\mathbf{I}_M)$, where each entry of $\mathbf{h}_k$ is assumed to be an i.i.d. Gaussian random variable with zero mean and unit variance. For the correlated Rayleigh fading channel, we consider a uniform linear array (ULA) at the BS modeled by the exponential correlation model. $\mathbf{h}_k$ is modeled as $\mathbf{h}_k\sim\mathcal{CN}(\mathbf{0},\mathbf{C}_k)$, where $\mathbf{C}_k$ is the spatial correlation matrix. The element at the ath row and the bth column of $\mathbf{C}_k$ is expressed as $[\mathbf{C}_k]_{a,b}=\rho^{-|b-a|}e^{j\delta_k(b-a)}$, where $0\le\rho\le1$ is the correlation coefficient between adjacent antennas and $-\pi\le\delta_k<\pi$ is the angle-of-arrival from the kth user to BS. In this work, perfect CSI is assumed to be available at the receiver. The QPSK and 16-QAM modulation are adopted. Unless otherwise stated, the number of users in each group is set to be $K_g=2$ for the proposed GW-EP/AGW-EP methods. The turbo code with code rate 1/2 is considered. $T_{\max}$ is set to be 6 for the iterative detectors. The MMSE, LAMA, MRC, CG, and GMP detectors are provided as baselines for comparison.

B. Simulation Results

Fig. 4 illustrates the BER performance of different detectors as a function of antenna cluster number C for the star architecture with $M=96$, $K=12$, and 16-QAM. For the spatially uncorrelated channel, Fig. 4(a) shows that all decentralized detectors (i.e., $C>1$) suffer BER performance loss compared with their centralized counterparts (i.e., $C=1$). This can be interpreted by the fact that the global information sharing is limited and each cluster only accesses local CSI and local received data. With the increase of C, the BER performance of decentralized linear methods deteriorates drastically compared with nonlinear algorithms. As expected, the proposed GW-EP detector achieves the best BER performance. The performance gap between the GW-EP and AGW-EP methods is almost indistinguishable.[5] For the spatially correlated channel, similar trends are also observed in Fig. 4(b).

Figs. 5 and 6 illustrate the BER performance comparison of different decentralized detectors for the star architecture when $M=96$ and $K=12$. For the spatially uncorrelated channel, it can be seen from Fig. 5(a) that the GW-EP detector outperforms its counterparts, especially for the scenarios with large C and high modulation order. For $C=4$ and 16-QAM, the proposed GW-WP detector obtains performance gains of

[5]For clarity, the performance of the AGW-EP detector with QPSK modulation scheme is not plotted in Figs. 5 and 6.

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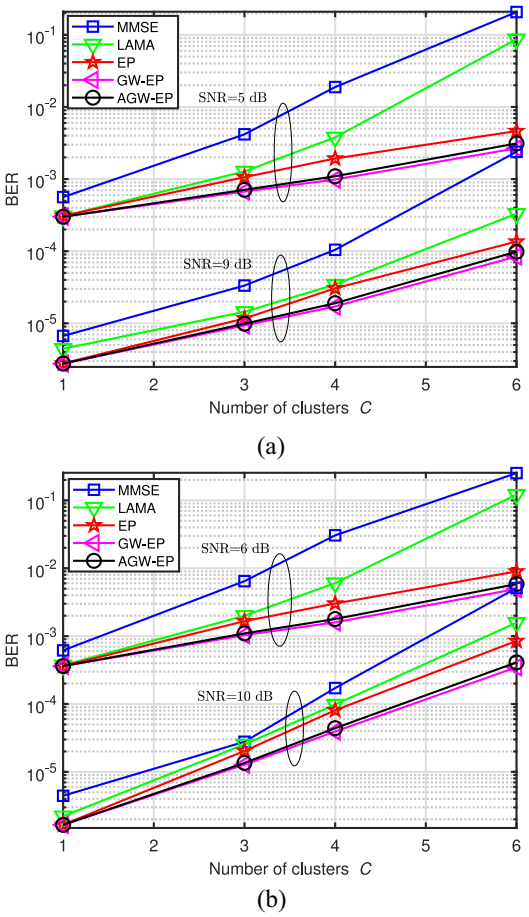


Fig. 4. BER performance of different detectors as a function of clusters number C with different SNR ($M = 96$, $K = 12$, and 16-QAM). (a) Uncorrelated. (b) Correlated, $\rho = 0.4$.

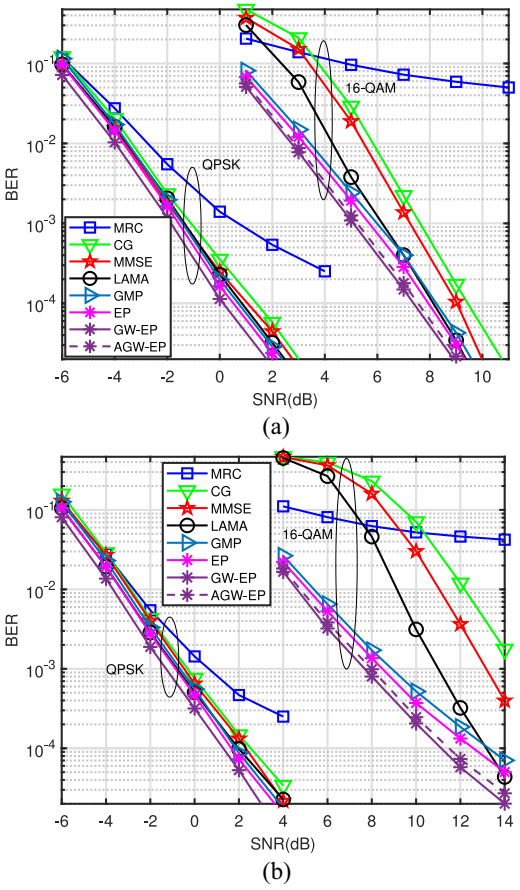


Fig. 5. BER performance comparison of the decentralized GW-EP detector with other recently proposed methods for uncorrelated channel ($M = 96$, $K = 12$, and $\rho = 0$). (a) $C = 4$, Uncorrelated. (b) $C = 8$, Uncorrelated.

about 0.5, 0.7, 1, 2, and 2.4 dB, respectively, over the conventional EP, GMP, LAMA, MMSE, and CG counterparts at BER $= 10^{-3}$. The performance loss of the AGW-EP detector is almost negligible compared with the GW-EP method. The results in Fig. 5(b) show that the performance gains of the GW-EP method are more pronounced when $C = 8$. Similar phenomena are also observed for the spatially correlated channel with $\rho = 0.4$, and the GW-EP method achieves the best BER performance as illustrated in Fig. 6(a) and (b).

Fig. 7 depicts the BER performance of the GW-EP and AGW-EP detectors with different values of K_g for the star architecture when $M = 64$ and $K = 8$. For the uncorrelated channel, Fig. 7 shows that the BER performance improves significantly with the increased K_g, especially for the star architecture with a large number of antenna clusters. For $C = 8$, the proposed GW-EP method with $K_g = 2$ and $K_g = 4$ yield about 1.2 and 3-dB gains at BER $= 10^{-4}$, respectively, over the conventional one (i.e., $K_g = 1$). This can be explained as follows: the proposed GW-EP method can fully exploit the groupwise information in each local cluster, whereas the conventional one fails to share the information among clusters. In addition, the proposed AGW-EP detector only suffers marginal performance loss, i.e., about 0.2 dB for $K_g = 2$ and 0.3 dB for $K_g = 4$ at BER $= 10^{-4}$, compared with the GW-EP detector.

Fig. 8 shows the BER performance of the GW-EP detector for the spatially correlated channel with different correlation coefficients ρ when $M = 64$, $K = 8$, and $C = 4$. It is shown that the proposed groupwise scheme exhibits great resilience to the channel correlation. For example, the performance loss of the GW-EP detector with $\rho = 0.5$ and $\rho = 0.6$ is about 0.4 and 0.7 dB compared to that with $\rho = 0.4$ at BER $= 10^{-4}$. In addition, the proposed GW-EP detector achieves greater performance gains over the conventional EP method with the increase of ρ. The BER performance gaps between $K_g = 4$ and $K_g = 1$ are approximately 0.6, 1.2, 2.2, and 5 dB at BER $= 10^{-3}$, respectively, for $\rho = 0$, $\rho = 0.4$, $\rho = 0.5$, and $\rho = 0.6$.

Fig. 9 shows the convergence of the proposed GW-EP method for the star architecture by tracking the value of $v_{f_c \to \mathbf{x}}$ [cf. (17)] in the spatially correlated channel when SNR $= 8$ dB, $M = 64$, $K = 8$, and $\rho = 0.4$. It is observed that for both $C = 4$ and $C = 8$, the proposed GW-EP algorithm converges faster with the increase of K_g. For $K_g = 4$, the GW-EP algorithm requires three iterations to converge, while $K_g = 1$ and $K_g = 2$ take about five iterations. The gap between $v_{f_c \to \mathbf{x}}$ and σ_n^2 decreases as K_g increases, which demonstrates the effectiveness of the proposed algorithm. In addition, as the antenna cluster increases, the value of $v_{f_c \to \mathbf{x}}$ in the proposed algorithm is far away from σ_n^2 due to the limited sharing of global information, which corroborates the BER analysis results in Fig. 4.

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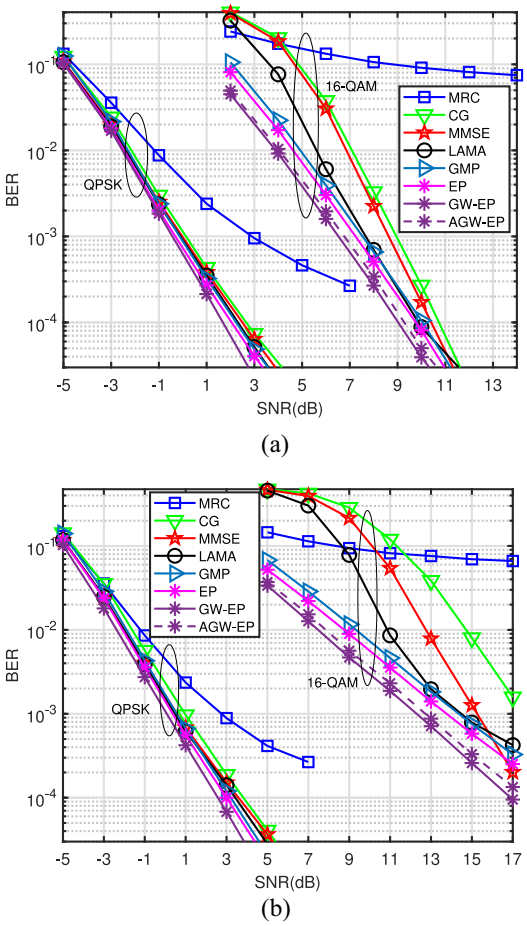


Fig. 6. BER performance comparison of the decentralized GW-EP detector with other recently proposed methods for correlated channel ($M = 96$, $K = 12$, and $\rho = 0.4$). (a) $C = 4$, Correlated, $\rho = 0.4$. (b) $C = 8$, Correlated, $\rho = 0.4$.

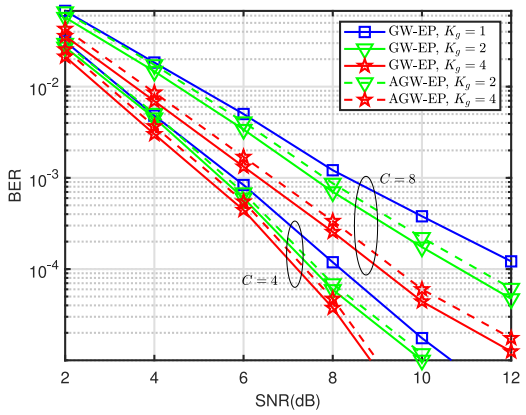


Fig. 7. BER performance of the GW-EP and AGW-EP detectors with different K_g for the star architecture ($M = 64$, $K = 8$, $\rho = 0$, and 16-QAM).

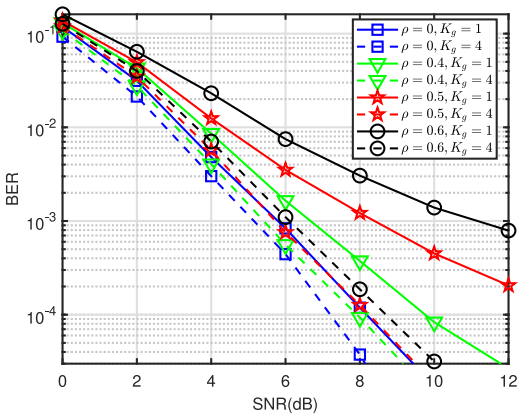


Fig. 8. BER performance of the GW-EP detector for the spatially correlated channel with different ρ ($M = 64$, $K = 8$, $C = 4$, and 16-QAM).

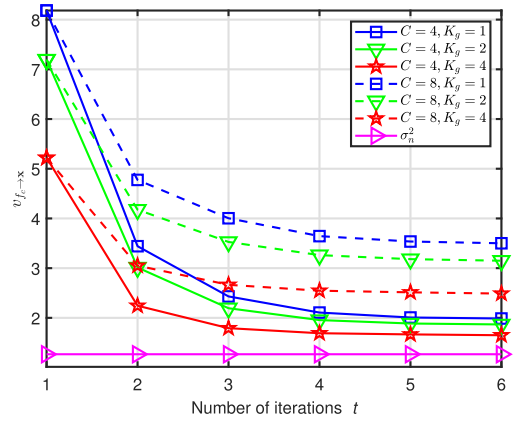


Fig. 9. State evolution of $v_{f_c \to \mathbf{x}}$ as a function of the number of iterations t for the star architecture ($M = 64$, $K = 8$, $\rho = 0.4$, and 16-QAM).

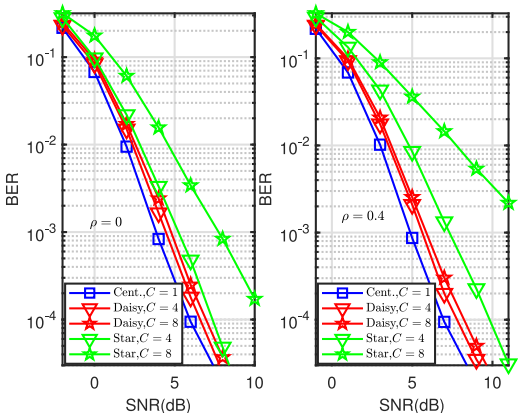


Fig. 10. BER performance of the GW-EP detector for the daisy-chain and star architectures ($M = 96$, $K = 12$, $K_g = 2$, and 16-QAM).

Fig. 10 illustrates the BER performance of the proposed GW-EP detector for both DBP architectures with $M = 96$, $K = 12$, $K_g = 2$, and 16-QAM. The performance of the centralized architecture is plotted as a benchmark. For both spatially uncorrelated and correlated channels, it is shown that the performance of the star architecture is severely degraded with the increase of antenna clusters. In contrast, the daisy-chain structure suffers less performance degradation because part of the information is shared between adjacent clusters, and its performance is close to that of the centralized scheme.

Fig. 11 illustrates the BER performance of the proposed GW-EP and AGW-EP detectors for the daisy-chain architecture when $M = 64$, $K = 8$, and $\rho = 0.4$. As expected, the proposed GW-EP detector obtains greater performance gains than the conventional EP method. The performance gap between the GW-EP and the AGW-EP detector is less than

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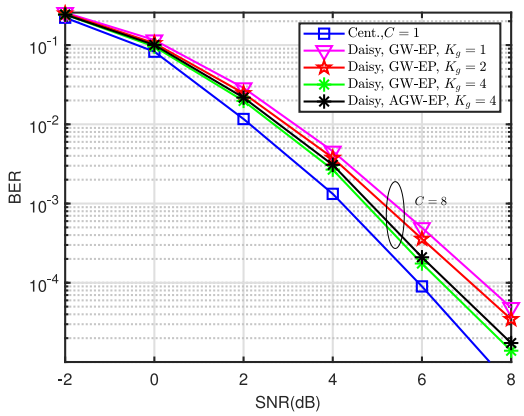


Fig. 11. BER performance of the GW-EP and AGW-EP detectors for the daisy-chain architecture ($M = 64$, $K = 8$, $\rho = 0.4$, and 16-QAM).

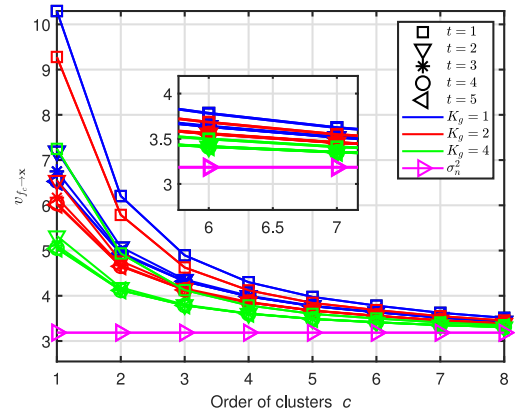


Fig. 12. State evolution of $v_{f_c\to\mathbf{x}}$ as a function of the order of antenna clusters c for the daisy-chain architecture ($M = 64$, $K = 8$, $C = 8$, $\rho = 0.4$, and 16-QAM).

0.2 dB at BER $= 10^{-4}$, which demonstrates the efficacy of the proposed AGW-EP method for the daisy-chain architecture.

Fig. 12 depicts the value of $v_{f_c\to\mathbf{x}}$ of the proposed GW-EP method for the daisy-chain architecture with the increasing order of antenna clusters c when SNR $= 4$ dB, $M = 64$, $K = 8$, $C = 8$, and $\rho = 0.4$. It is shown that the gap between $v_{l,f\to\mathbf{x}}$ and σ_n^2 decreases as the order of antenna clusters c increases. The number of required iterations of the proposed GW-EP method to achieve numerical stability decreases with the increase of c. This is attributed to the fact that the a priori information from the preceding antenna clusters can be adopted by the current cluster. Similar to the phenomenon in the star architecture, the gap between $v_{f_c\to\mathbf{x}}$ and σ_n^2 decreases as K_g increases owing to the full use of groupwise information.

Fig. 13 shows the computational complexity comparison of different detectors against the number of users K when $N = 96$, $C = 4$, and $K_g = 2$. It is shown that the computational complexity of the proposed GW-EP detector scales linearly with the number of users, as opposed to the cubic order of the MMSE method. As expected, the approximate process (AGW-EP) can further reduce the computational complexity of the GW-EP detector, especially for the high-order modulation (16QAM).

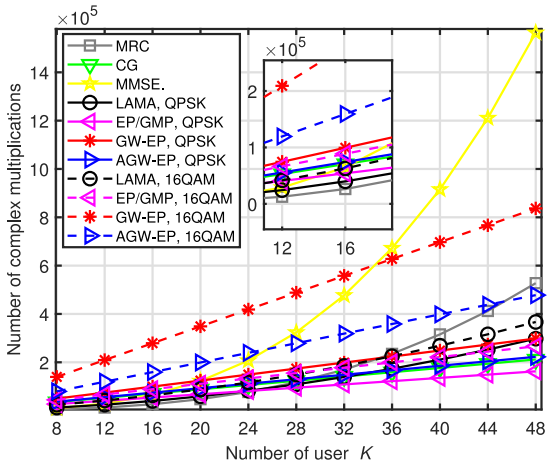


Fig. 13. Complexity comparison of different receivers with different number of users K ($N = 96$, $C = 4$, and $K_g = 2$).

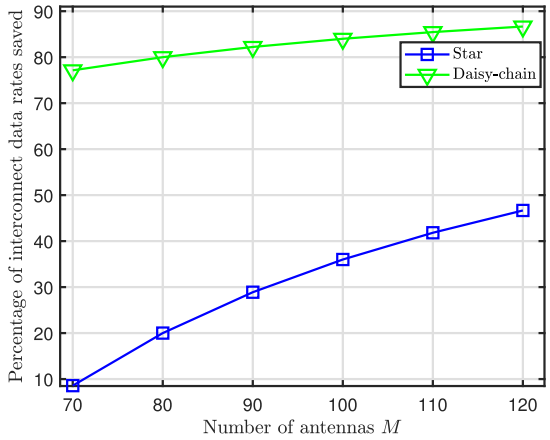


Fig. 14. Percentage of interconnect data rates saved for the proposed GW-EP algorithm over centralized implementation as a function of the number of antennas M ($K = 8$, $C = 4$, and $K_g = 2$).

Fig. 14 presents the percentage of interconnect data rates saved for the proposed GW-EP algorithm over centralized implementation as a function of the number of antennas M when $K = 8$, and $C = 4$. It is observed that the reduced interconnect data rates of the proposed method for star and daisy-chain architectures increases with M, which enables the scalability required for MU-MIMO networks with a large number of BS antennas.

V. Conclusion

In this article, we have focused on the massive MU-MIMO technology to support the requirement of massive connectivity in IoT networks. We proposed a novel decentralized group-wise detection paradigm and applied it to the EP-based DBP equalization of the uplink massive MU-MIMO system. A nonlinear message fusion rule was designed based on the product principle of multivariate Gaussian PDF for star and daisy-chain architectures, respectively. To reduce the computational complexity of the GW-EP method, we proposed an AGW-EP algorithm based on the most reliable constellation vectors.

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$$
\begin{aligned}
v^t_{f_{c,n}\to\mathbf{x}_g} &= \sigma_n^2 + \mathbf{e}_{c,n}^{\mathrm{T}}\left(\sum_{g'\neq g}\mathbf{H}_{c,g'}\mathbf{V}^t_{\mathbf{x}_{g'}\to f_{c,n}}\mathbf{H}^{\mathrm{H}}_{c,g'}\right)\mathbf{e}_{c,n}\\
&= \sigma_n^2 + \mathbf{e}_{c,n}^{\mathrm{T}}\left(\sum_{g'\neq g}\mathbf{H}_{c,g'}\left(\left(\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\right)^{-1} - \frac{\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}}\right)^{-1}\mathbf{H}^{\mathrm{H}}_{c,g'}\right)\mathbf{e}_{c,n}\\
&\overset{(a)}{=} \sigma_n^2 + \mathbf{e}_{c,n}^{\mathrm{T}}\left(\sum_{g'\neq g}\mathbf{H}_{c,g'}\left(\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}} + \frac{\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right)\mathbf{H}^{\mathrm{H}}_{c,g'}\right)\mathbf{e}_{c,n}\\
&= \sigma_n^2 + \left(\sum_{g'\neq g}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\right) + \left(\sum_{g'\neq g}\frac{\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right)\\
&= \sigma_n^2 + \sum_{g'\neq g}\left(\frac{\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\left(v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\right)}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right.\\
&\quad\left. + \frac{\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right)\\
&= \sigma_n^2 + \left(\sum_{g'}\frac{\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right) - \frac{\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_g}\mathbf{H}^{\mathrm{H}}_{c,g}\mathbf{e}_{c,n}v^{t-1}_{f_{c,n}\to\mathbf{x}_g}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_g} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_g}\mathbf{H}^{\mathrm{H}}_{c,g}\mathbf{e}_{c,n}}\\
&= \sigma_n^2 + \sum_{g'}\psi^t_{c,n,g'}\varphi^t_{c,n,g'} - \psi^t_{c,n,g}\varphi^t_{c,n,g} \qquad (36)\\
u^t_{f_{c,n}\to\mathbf{x}_g} &= y_{c,n} - \mathbf{e}_{c,n}^{\mathrm{T}}\sum_{g'\neq g}\mathbf{H}_{c,g'}\mathbf{u}^t_{\mathbf{x}_{g'}\to f_{c,n}}\\
&= y_{c,n} - \mathbf{e}_{c,n}^{\mathrm{T}}\sum_{g'\neq g}\mathbf{H}_{c,g'}\left(\mathbf{V}^t_{\mathbf{x}_{g'}\to f_{c,n}}\left(\left(\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\right)^{-1}\widetilde{\mathbf{u}}^t_{c,\mathbf{x}_{g'}} - \frac{u^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right)\right)\\
&= y_{c,n} - \mathbf{e}_{c,n}^{\mathrm{T}}\sum_{g'\neq g}\mathbf{H}_{c,g'}\left(\left(\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}} + \frac{\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right)\left(\left(\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\right)^{-1}\widetilde{\mathbf{u}}^t_{c,\mathbf{x}_{g'}} - \frac{u^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right)\right)\\
&= y_{c,n} - \sum_{g'\neq g}\left(\left(\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{u}}^t_{c,\mathbf{x}_{g'}} - \frac{u^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}}\mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}\right)\frac{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}}{v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}} - \mathbf{e}_{c,n}^{\mathrm{T}}\mathbf{H}_{c,g'}\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}}\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}}\right)\\
&= y_{c,n} - \sum_{g'}\phi^t_{c,n,g'}\varphi^t_{c,n,g'} + \phi^t_{c,n,g}\varphi^t_{c,n,g} \qquad (37)
\end{aligned}
$$

The state evolution illustrates that the proposed decentralized GW-EP achieves faster convergence than the conventional one. Simulation results demonstrate that the proposed GW-EP and AGW-EP detectors achieve significant performance gains over the conventional decentralized methods of the star architecture, especially for scenarios with lower BS-to-user ratios and/or spatially correlated channels. For the daisy-chain architecture, the proposed detectors achieve comparable performance to that of the centralized detector. The proposed scheme delivers a flexible tradeoff between the BER performance and interconnect bandwidth requirements, which renders it suitable for implementation in practical systems. A rigorous BER performance analysis of the proposed methods and decentralized channel estimation algorithm are left for a future work.

APPENDIX
DERIVATION OF (18) AND (19)

By substituting (14) and (15) into (16) and (17), we obtain (36) and (37), shown at the top of the page, where (a) holds based on the Sherman–Morrison formula [40]

$$\left(\mathbf{A}+\mathbf{x}\mathbf{y}^{\mathrm{T}}\right)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{x}\mathbf{y}^{\mathrm{T}}\mathbf{A}^{-1}}{1+\mathbf{y}^{\mathrm{T}}\mathbf{A}^{-1}\mathbf{x}} \qquad (35)$$

where $\mathbf{A} = (\widetilde{\mathbf{V}}^t_{c,\mathbf{x}_{g'}})^{-1}$, $\mathbf{x} = ([\mathbf{H}^{\mathrm{H}}_{c,g'}\mathbf{e}_{c,n}]/[v^{t-1}_{f_{c,n}\to\mathbf{x}_{g'}}])$, and $\mathbf{y} = \mathbf{H}^{\mathrm{T}}_{c,g'}\mathbf{e}_{c,n}$.

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